MSG No. 123.348 P_c4/mmm [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/mmm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m'mm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]`

Table 3: Wyckoff site: `2c`, site symmetry: `4/mmm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 4: Wyckoff site: `2d`, site symmetry: `4/m'mm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]`

Table 5: Wyckoff site: `4e`, site symmetry: `m'mm.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,12,13,20,21,25,30]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[2,3,15,16,23,24,26,27]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[4,5,9,14,17,22,28,29]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[7,8,10,11,18,19,31,32]`

Table 6: Wyckoff site: `4f`, site symmetry: `mmm.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,6,9,12,13,14]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,7,8,10,11,15,16]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,20,21,22,25,28,29,30]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[18,19,23,24,26,27,31,32]`

Table 7: Wyckoff site: `4g`, site symmetry: `4mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6,12,13,15,16]`
2	$[0,\ 0,\ -z]$	`[4,5,7,8,9,10,11,14]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[17,18,19,22,28,29,31,32]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[20,21,23,24,25,26,27,30]`

Table 8: Wyckoff site: `4h`, site symmetry: `4mm`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,7,8,9,10,11,14]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,18,19,22,28,29,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[20,21,23,24,25,26,27,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,12,13]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,15,16]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8,10,11]`
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,22,28,29]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[18,19,31,32]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[20,21,25,30]`
8	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[23,24,26,27]`

Table 10: Wyckoff site: `8j`, site symmetry: `m.2m`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7,14,16]`
2	$[-x,\ x,\ 0]$	`[2,5,11,12]`
3	$[x,\ -x,\ 0]$	`[3,4,10,13]`
4	$[-x,\ -x,\ 0]$	`[6,8,9,15]`
5	$\left[x,\ x,\ \frac{1}{2}\right]$	`[17,23,30,32]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[18,21,27,28]`
7	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[19,20,26,29]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[22,24,25,31]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,16,23,30]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,12,21,27]`

continued ...

Table 11

No.	position	mapping
3	$\left[x,\, -x,\, \frac{1}{4}\right]$	[3,13,20,26]
4	$\left[x,\, -x,\, \frac{3}{4}\right]$	[4,10,19,29]
5	$\left[-x,\, x,\, \frac{3}{4}\right]$	[5,11,18,28]
6	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[6,15,24,25]
7	$\left[x,\, x,\, \frac{3}{4}\right]$	[7,14,17,32]
8	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[8,9,22,31]

Table 12: Wyckoff site: 8l, site symmetry: m2m.

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,4,13,14]
2	$[0,\, x,\, 0]$	[2,7,11,16]
3	$[0,\, -x,\, 0]$	[3,8,10,15]
4	$[-x,\, 0,\, 0]$	[5,6,9,12]
5	$\left[x,\, 0,\, \frac{1}{2}\right]$	[17,20,29,30]
6	$\left[0,\, x,\, \frac{1}{2}\right]$	[18,23,27,32]
7	$\left[0,\, -x,\, \frac{1}{2}\right]$	[19,24,26,31]
8	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[21,22,25,28]

Table 13: Wyckoff site: 8m, site symmetry: m'2'm.

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{4}\right]$	[1,13,20,30]
2	$\left[0,\, x,\, \frac{1}{4}\right]$	[2,16,23,27]
3	$\left[0,\, -x,\, \frac{1}{4}\right]$	[3,15,24,26]
4	$\left[x,\, 0,\, \frac{3}{4}\right]$	[4,14,17,29]
5	$\left[-x,\, 0,\, \frac{3}{4}\right]$	[5,9,22,28]
6	$\left[-x,\, 0,\, \frac{1}{4}\right]$	[6,12,21,25]
7	$\left[0,\, x,\, \frac{3}{4}\right]$	[7,11,18,32]
8	$\left[0,\, -x,\, \frac{3}{4}\right]$	[8,10,19,31]

Table 14: Wyckoff site: 8n, site symmetry: m2m.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,4,13,14]
2	$\left[\frac{1}{2},\, x,\, 0\right]$	[2,7,11,16]
3	$\left[\frac{1}{2},\, -x,\, 0\right]$	[3,8,10,15]
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	[5,6,9,12]
5	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[17,20,29,30]
6	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[18,23,27,32]
7	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[19,24,26,31]
8	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[21,22,25,28]

Table 15: Wyckoff site: 8o, site symmetry: m'2'm.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,13,20,30]
2	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[2,16,23,27]
3	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,15,24,26]
4	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[4,14,17,29]
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,9,22,28]
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[6,12,21,25]
7	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,11,18,32]
8	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[8,10,19,31]

Table 16: Wyckoff site: 16p, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-y,\ x,\ 0]$	[2,11]
3	$[y,\ -x,\ 0]$	[3,10]
4	$[x,\ -y,\ 0]$	[4,13]
5	$[-x,\ y,\ 0]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$[y,\ x,\ 0]$	[7,16]
8	$[-y,\ -x,\ 0]$	[8,15]
9	$\left[x,\ y,\ \frac{1}{2}\right]$	[17,30]
10	$\left[-y,\ x,\ \frac{1}{2}\right]$	[18,27]
11	$\left[y,\ -x,\ \frac{1}{2}\right]$	[19,26]
12	$\left[x,\ -y,\ \frac{1}{2}\right]$	[20,29]
13	$\left[-x,\ y,\ \frac{1}{2}\right]$	[21,28]
14	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[22,25]
15	$\left[y,\ x,\ \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[24,31]

Table 17: Wyckoff site: 16q, site symmetry: m'..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,30]
2	$\left[-y,\ x,\ \frac{1}{4}\right]$	[2,27]
3	$\left[y,\ -x,\ \frac{1}{4}\right]$	[3,26]
4	$\left[x,\ -y,\ \frac{3}{4}\right]$	[4,29]
5	$\left[-x,\ y,\ \frac{3}{4}\right]$	[5,28]
6	$\left[-x,\ -y,\ \frac{1}{4}\right]$	[6,25]
7	$\left[y,\ x,\ \frac{3}{4}\right]$	[7,32]
8	$\left[-y,\ -x,\ \frac{3}{4}\right]$	[8,31]
9	$\left[-x,\ -y,\ \frac{3}{4}\right]$	[9,22]
10	$\left[y,\ -x,\ \frac{3}{4}\right]$	[10,19]
11	$\left[-y,\ x,\ \frac{3}{4}\right]$	[11,18]
12	$\left[-x,\ y,\ \frac{1}{4}\right]$	[12,21]

continued ...

Table 17

No.	position	mapping
13	$\left[x,\, -y,\, \frac{1}{4}\right]$	[13,20]
14	$\left[x,\, y,\, \frac{3}{4}\right]$	[14,17]
15	$\left[-y,\, -x,\, \frac{1}{4}\right]$	[15,24]
16	$\left[y,\, x,\, \frac{1}{4}\right]$	[16,23]

Table 18: Wyckoff site: 16r, site symmetry: ..m

No.	position	mapping
1	$[x,\, x,\, z]$	[1,16]
2	$[-x,\, x,\, z]$	[2,12]
3	$[x,\, -x,\, z]$	[3,13]
4	$[x,\, -x,\, -z]$	[4,10]
5	$[-x,\, x,\, -z]$	[5,11]
6	$[-x,\, -x,\, z]$	[6,15]
7	$[x,\, x,\, -z]$	[7,14]
8	$[-x,\, -x,\, -z]$	[8,9]
9	$\left[x,\, x,\, z+\frac{1}{2}\right]$	[17,32]
10	$\left[-x,\, x,\, z+\frac{1}{2}\right]$	[18,28]
11	$\left[x,\, -x,\, z+\frac{1}{2}\right]$	[19,29]
12	$\left[x,\, -x,\, \frac{1}{2}-z\right]$	[20,26]
13	$\left[-x,\, x,\, \frac{1}{2}-z\right]$	[21,27]
14	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	[22,31]
15	$\left[x,\, x,\, \frac{1}{2}-z\right]$	[23,30]
16	$\left[-x,\, -x,\, \frac{1}{2}-z\right]$	[24,25]

Table 19: Wyckoff site: 16s, site symmetry: .m.

No.	position	mapping
1	$[x,\, 0,\, z]$	[1,13]
2	$[0,\, x,\, z]$	[2,16]
3	$[0,\, -x,\, z]$	[3,15]
4	$[x,\, 0,\, -z]$	[4,14]
5	$[-x,\, 0,\, -z]$	[5,9]
6	$[-x,\, 0,\, z]$	[6,12]
7	$[0,\, x,\, -z]$	[7,11]
8	$[0,\, -x,\, -z]$	[8,10]
9	$\left[x,\, 0,\, z+\frac{1}{2}\right]$	[17,29]
10	$\left[0,\, x,\, z+\frac{1}{2}\right]$	[18,32]
11	$\left[0,\, -x,\, z+\frac{1}{2}\right]$	[19,31]
12	$\left[x,\, 0,\, \frac{1}{2}-z\right]$	[20,30]
13	$\left[-x,\, 0,\, \frac{1}{2}-z\right]$	[21,25]
14	$\left[-x,\, 0,\, z+\frac{1}{2}\right]$	[22,28]
15	$\left[0,\, x,\, \frac{1}{2}-z\right]$	[23,27]
16	$\left[0,\, -x,\, \frac{1}{2}-z\right]$	[24,26]

Table 20: Wyckoff site: `16t`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,13]`
2	$\left[\frac{1}{2},\ x,\ z\right]$	`[2,16]`
3	$\left[\frac{1}{2},\ -x,\ z\right]$	`[3,15]`
4	$\left[x,\ \frac{1}{2},\ -z\right]$	`[4,14]`
5	$\left[-x,\ \frac{1}{2},\ -z\right]$	`[5,9]`
6	$\left[-x,\ \frac{1}{2},\ z\right]$	`[6,12]`
7	$\left[\frac{1}{2},\ x,\ -z\right]$	`[7,11]`
8	$\left[\frac{1}{2},\ -x,\ -z\right]$	`[8,10]`
9	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,29]`
10	$\left[\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[18,32]`
11	$\left[\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	`[19,31]`
12	$\left[x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[20,30]`
13	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[21,25]`
14	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[22,28]`
15	$\left[\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	`[23,27]`
16	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[24,26]`

Table 21: Wyckoff site: `32u`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x,\ z]$	`[2]`
3	$[y,\ -x,\ z]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[-x,\ y,\ -z]$	`[5]`
6	$[-x,\ -y,\ z]$	`[6]`
7	$[y,\ x,\ -z]$	`[7]`
8	$[-y,\ -x,\ -z]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$[y,\ -x,\ -z]$	`[10]`
11	$[-y,\ x,\ -z]$	`[11]`
12	$[-x,\ y,\ z]$	`[12]`
13	$[x,\ -y,\ z]$	`[13]`
14	$[x,\ y,\ -z]$	`[14]`
15	$[-y,\ -x,\ z]$	`[15]`
16	$[y,\ x,\ z]$	`[16]`
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	`[17]`
18	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[18]`
19	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[19]`
20	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[20]`
21	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[21]`
22	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[22]`
23	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[23]`
24	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[24]`
25	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	`[25]`

continued ...

Table 21

No.	position	mapping
26	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[30]
31	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[32]